Tesla vs GameStop Stock & Revenue Analysis

Project Overview

This project analyzes the relationship between **company revenue growth and stock market performance** by comparing two companies with very different market behaviors: **Tesla (TSLA)** and **GameStop (GME)**.

Using Python-based data analytics tools, historical stock price data and company revenue data were collected, cleaned, and visualized to identify patterns between **financial performance and investor market behavior**.

The goal of the analysis was to determine whether stock price trends align with **fundamental business performance** or whether external factors such as market sentiment influence price movements.

Data & Methodology

Data Sources

- Yahoo Finance (Stock price data)
- Macrotrends (Company revenue data)

Tools & Technologies

- Python
- Pandas
- BeautifulSoup (Web scraping)
- yfinance API
- Plotly (Interactive visualization)
- Jupyter Notebook

Analysis Workflow

1. Extract historical stock price data for Tesla and GameStop
2. Scrape quarterly revenue data from financial data sources
3. Clean and transform datasets for consistency
4. Build visualization dashboards comparing stock price vs revenue
5. Analyze trends and generate business insights

Key Visualizations

Figure 1 – Tesla Stock Price vs Revenue Trend

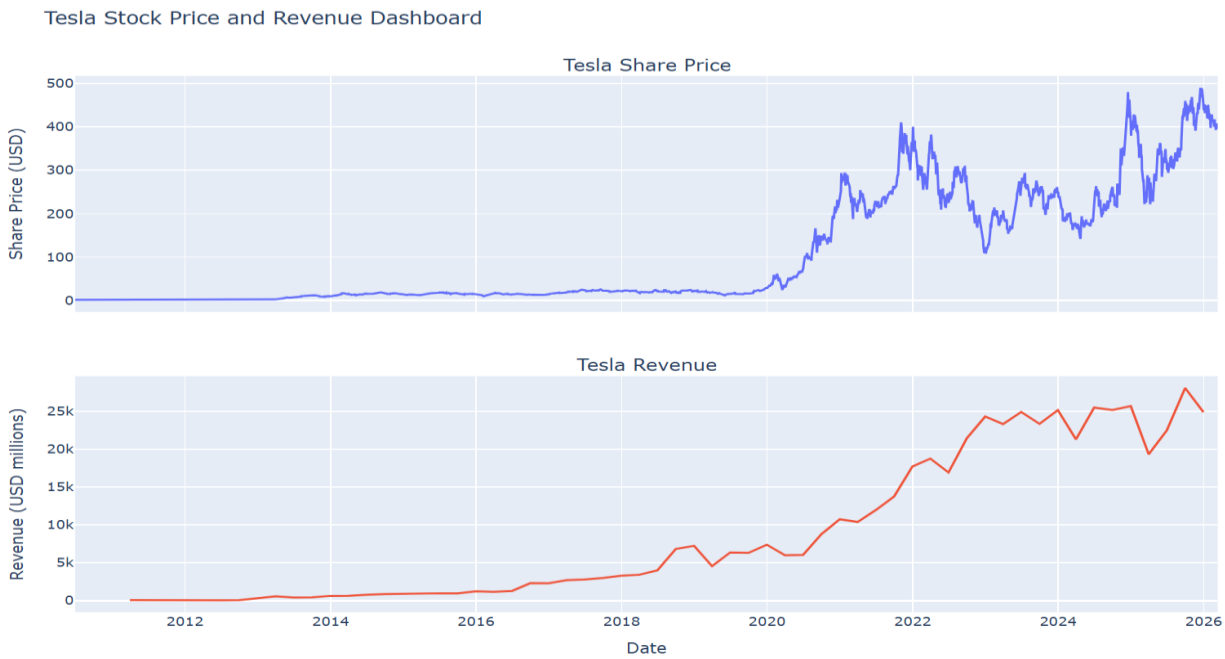
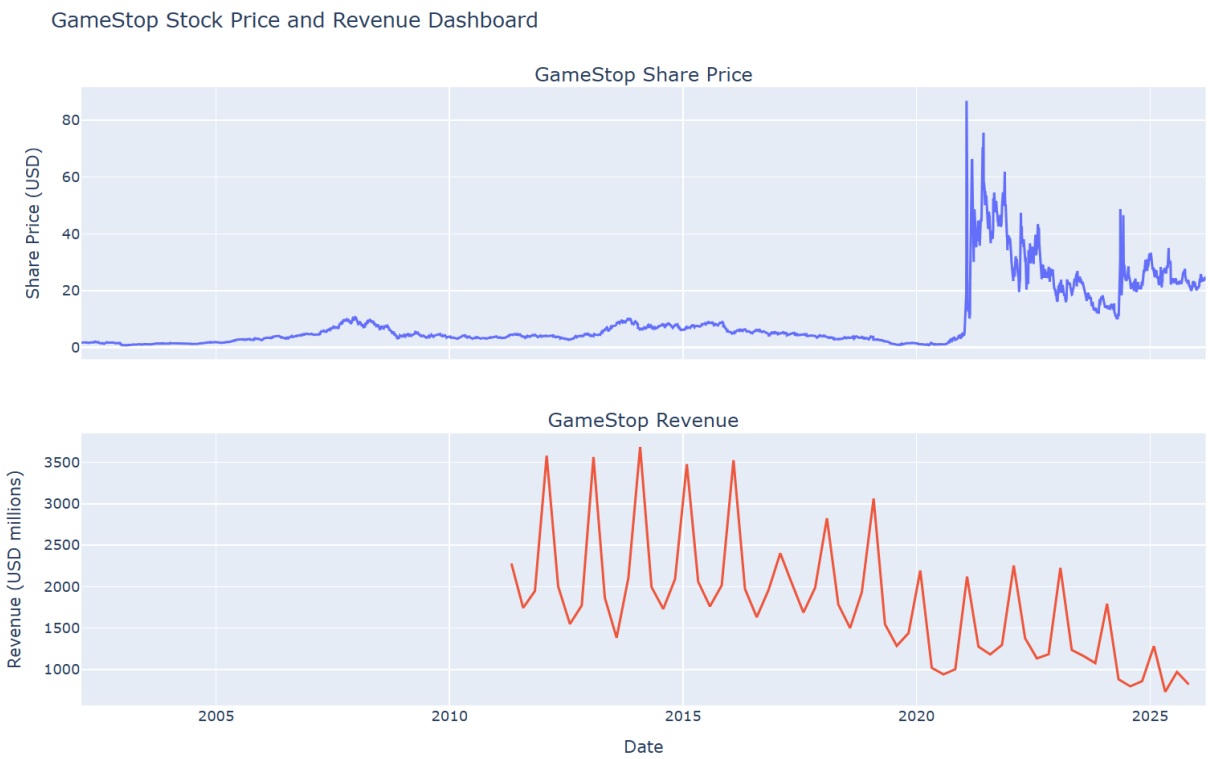


Figure 2 – GameStop Stock Price vs Revenue Trend



Key Insights

Tesla

- Tesla demonstrates **consistent revenue growth over time**.
- Stock price trends generally align with revenue growth, indicating **strong investor confidence linked to business performance**.
- The company's innovation and market expansion have contributed to **long-term valuation growth**.

GameStop

- GameStop's revenue trend remained relatively stable with moderate decline periods.
- However, its stock price experienced **extreme volatility** that was not directly tied to company fundamentals.
- Market events such as the **retail investor trading surge in 2021** significantly impacted its stock price behavior.

Business Interpretation

This analysis highlights an important concept in financial markets:

- **Fundamental growth companies** (like Tesla) tend to show stronger correlation between revenue performance and stock valuation.
- **Market-driven stocks** can experience price movements heavily influenced by **investor sentiment and speculation rather than financial performance**.

Understanding this difference is critical for investors, analysts, and financial strategists when evaluating market opportunities.

Conclusion

The project demonstrates how data analytics can be used to evaluate relationships between **company performance and market valuation**.

Through data extraction, cleaning, and visualization, the analysis shows that while financial performance often influences stock prices, external market dynamics can also create significant price volatility.

This project showcases practical **data analytics skills including data extraction, web scraping, data cleaning, and interactive visualization**, which are widely used in modern financial and business analytics.